SARDAR PATEL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

LAN-BASED SECURE CHATTING APPLICATION

A self-hosted local communication platform with secure messaging, PostgreSQL persistence, and lightweight data-science modules.

Aryan Patil
2024300169

Malhar Nimbalkar
2024300150

Amogh Sirsikar
2024300245

GROUP 18
PHASE 1

Problem Statement

LOCAL

Communication boundary

PRIVATE

Data ownership

SECURE

Transport + storage

SMART

PDS-assisted chat

THE PROBLEM

Typical messaging platforms depend on external infrastructure and are designed for Internet-scale communication. For a classroom, office LAN or isolated lab, that can add unnecessary dependency and reduce local control over data.

WHY IT MATTERS

Users need fast local messaging, file exchange and searchable history while keeping communication inside the trusted network boundary. Security research also shows that secure messaging must balance confidentiality with usability.

Project Objectives and Scope

OBJECTIVES

01 USER & AUTH

Registration/login, user identity, online status and basic access control.

02 REAL-TIME CHAT

One-to-one and group messaging over the local network.

03 PERSISTENCE

PostgreSQL stores users, conversations, messages, timestamps and file metadata.

04 SECURITY

Authenticated connections, protected credentials and encrypted message transport.

05 SPAM / MISINFO

Python Logistic Regression classifies messages as likely spam / potentially misleading.

SCOPE / BOUNDARY

CORE DELIVERABLE

Secure LAN communication with Java clients/server, PostgreSQL persistence, authentication and encrypted transport.

SUPPORTING PDS

Logistic Regression for spam / potentially misleading-message screening and Linear Regression for a measurable system metric.

EXPLICIT LIMIT

Only Logistic Regression are proposed in the PDS module.

Literature Survey

Author / Year	Area	Key finding / relevance
Johansen et al., 2021	Secure messaging survey	Compared Signal, WhatsApp, Wire, Viber, Riot and Telegram across 21 security/usability properties; none was perfect. [1]
Dedeturk & Akay, 2020	Spam + Logistic Regression	Showed Logistic Regression can be an efficient real-time classifier for spam; used text features and public datasets. [2]

Author / Year	Paper / venue	Relevance to the project
Paterson, Scarlata & Truong, 2023	Three Lessons From Threema: Analysis of a Secure Messenger - USENIX Security 23	Cryptographic analysis found protocol attacks and highlighted the need for careful threat modelling and remediation. [6]
Kuros, J. F. and Ross, K. W.	Computer Networking : A Top Down Approach 8th Ed Pearson 2021	Provides Communication Foundation for Reliable Data Exchange using TCP/IP and Socket Programming. [5]

RELATED LAN SYSTEMS

- BeeBEEP: serverless LAN messenger with encrypted P2P chat, file transfer and message history.
- IP Messenger: lightweight serverless LAN messenger using TCP/UDP with RSA-2048 + AES-256 + SHA-256.

Gap Analysis

The project is not intended to replace Signal; it combines LAN-first architecture, relational persistence and lightweight PDS.

System	Network model	Security	Persistence	Limitation / gap
Signal / WhatsApp	Internet + servers	Strong E2E	Server / cloud	Not LAN-first; large-scale infrastructure.
Telegram	Internet + servers	E2E only in Secret Chat	Server-based	Security model differs by chat mode.
BeeBEEP	LAN / P2P	AES-based	Local history	No integrated PDS layer; different architecture.
IP Messenger	LAN / serverless	RSA + AES + SHA	Local logs	Lightweight utility; limited analytics/DB integration.
Our proposal	LAN + Java server	TLS + protected storage	PostgreSQL	Adds interpretable PDS modules + unified DB design.

IDENTIFIED GAP

A small, explainable, LAN-only academic system that integrates secure communication + PostgreSQL + PDS techniques in one demonstrable pipeline.

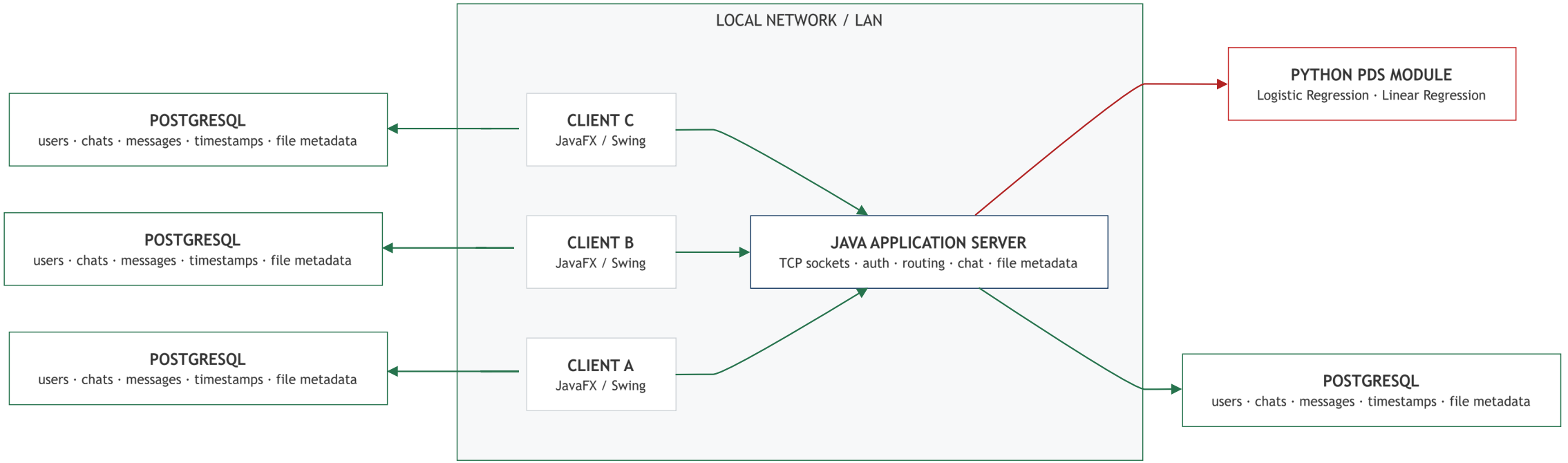
Mapping of Project with Subject Concepts

Each major module demonstrates a concept from the corresponding subject.

Subject	Concept applied	Where it appears in the project	Demonstrable outcome
Computer Networks	LAN communication; TCP sockets; client-server model; authenticated connections; transport security	Java clients communicate with the Java application server over the local network.	Message delivery, concurrent clients, LAN boundary and secure transport.
DBMS	Relational schema; CRUD; persistence; indexing; transactions / controlled DB access	PostgreSQL stores users, conversations, messages, timestamps and file metadata.	Persistent chat history, searchable data and measurable DB response time.
Software Engineering	Requirements; scope; architecture; planning; risk analysis; testing; validation	Project planning, modular workflow, acceptance criteria and risk/validation slide.	Traceable requirements, defined scope and testable success metrics.
PDS	Logistic Regression; Linear Regression; feature selection; classification/regression evaluation	Python module screens messages and estimates a measurable system metric.	Probability/label output plus MAE/RMSE evaluation on collected LAN observations.
PSOOP	Classes and objects; encapsulation; inheritance / polymorphism; modular OOP design	Java client, server and supporting classes organize chat, authentication and data-handling modules.	Reusable classes, modular components and object-oriented message / connection handling.
DSGT	Graphs; vertices and edges; paths; connectivity; relations	The LAN can be represented as a graph of clients and the server, with communication links as edges.	Network topology, connectivity and possible communication paths can be modelled and analysed.

Proposed Architecture

One LAN, one backend, four cooperating layers.



PDS Integration

We will be using Logistic Regression for spam detection or fake news detection

LOGISTIC REGRESSION

Message → probability → label

Use case: Spam / potentially misleading message screening

Output: P(spam) or P(misleading) + thresholded label

Important limitation: It flags patterns; it does NOT fact-check truth against the web.

Model: $p(y=1|x) = 1 / (1 + e^{-z})$

End-to-End Workflow

1 DISCOVER

Client identifies the LAN server.

2 AUTH

Credentials verified; session established.

3 SEND

Message sent through authenticated Java connection.

4 PERSIST

Server writes message + metadata to PostgreSQL.

5 PDS

Python module scores spam/misinfo or predicts a system metric.

6 DELIVER

Server routes message to recipient(s) and returns status.

Project Planning

A 14-week plan from requirements to demo.

Activity	Weeks
Requirements + literature	1–4
UI + LAN socket prototype	4–7
PostgreSQL schema + CRUD	5–8
Authentication + security	8–11
Chat + group messaging	9–12
Python Logistic Regression	11–13
Python Linear Regression	12–14
Integration + testing	13–14
Performance + security testing	14
Report + PPT + demo	14

TEAM RESPONSIBILITIES

ARYAN PATIL

Networking + Java server/client

MALHAR NIMBALKAR

DBMS + backend

AMOGH SIRSIRKAR

PDS + UI

Risks and Validation

Phase 1 defines measurable acceptance criteria before implementation begins.

Risk	Mitigation / validation
LAN discovery fails	Use fixed server IP / configurable discovery as fallback; test across multiple hosts.
Database bottleneck	Use indexed message tables, prepared statements and pagination for history.
Security claim is too broad	Clearly separate transport security, credential security and message-storage security.
Classifier false positives	Report probability + threshold; allow user feedback and evaluate precision/recall.
Linear model is weak	Use it only for a measurable short-term metric and report MAE/RMSE honestly.
Python–Java integration adds latency	Run inference locally and exchange compact JSON requests/responses.

References

Research papers, related systems and source material. Numbering used throughout the literature-survey slides.

[1] C. Johansen, A. Mujaj, H. Arshad, J. Noll, “The Snowden Phone: A Comparative Survey of Secure Instant Messaging Mobile Applications,” Security and Communication Networks, 2021. DOI: 10.1155/2021/9965573.

[2] B. K. Dedeturk, B. Akay, “Spam filtering using a logistic regression model trained by an artificial bee colony algorithm,” Applied Soft Computing, 91, 106229, 2020. DOI: 10.1016/j.asoc.2020.106229.

[3] BeeBEEP, “Free Office Messenger” - LAN/P2P messaging, AES-based encryption, file sharing and message history. Official project documentation, accessed 2026.

[4] IP Messenger, “LAN Message Communication Software” TCP/UDP, RSA-2048 + AES-256 + SHA-256, file transfer. Official documentation, accessed 2026.

[5] Kurose, J. F. and Ross, K. W. , Computer Networking : A Top Down Approach 8th Ed, Pearson 2021

[6] K. G. Paterson, M. Scarlata, K. T. Truong, “Three Lessons From Threema: Analysis of a Secure Messenger,” 32nd USENIX Security Symposium, 2023, pp. 1289–1306.